

PROJECT PLAN DOCUMENT

Project number	2
Project Title	Robert — AI Medical Assistant (Iteration 4)
Document	Project Plan
Creation date	April 21, 2026
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Client	NanoCare Health Services Private Limited

Brief problem statement

Electronic Health Records (EMR) contain critical patient information but are often unstructured and difficult to interpret efficiently in clinical contexts. Standard large language models lack personalization, clinical grounding, and strict privacy controls. This project aims to develop Robert, a GDPR-compliant, clinically aware AI assistant that provides personalized medical guidance by synthesizing EMR data using a SNOMED-grounded Retrieval-Augmented Generation (RAG) pipeline. The system introduces an intent-driven term extraction phase to drastically improve diagnostic accuracy, coupled with a robust anonymization layer to adhere to strict data privacy constraints.

Team Members

- **Venya Velmurugan** - Developing multilingual interface support and integrating SNOMED CT knowledge graphs within the system's reasoning pipeline.
- **Advait Dintakurti** - Designing the modular system architecture, FastAPI backend development, and implementing the Next.js backend-frontend integration framework.
- **Vikesh Kansal** - Implementing hybrid retrieval, SapBERT semantic mapping, and vector indexing.
- **Keshav Goel** - Working on retrieval ranking strategies, pipeline orchestration, evaluation mechanisms, and system validation testing.
- **Amish Goyal** - Developing EMR ingestion, data processing modules, context compaction, and implementing GDPR session compliance layers.

Team Communication

The team conducts weekly structured meetings with the client mentor (NanoCare Health Services Private Limited) every Wednesday at 4:00 pm to review progress, gather feedback, and align on priorities. Internal team coordination meetings occur at least twice weekly for technical synchronization and task allocation. Primary communication channels include WhatsApp

for daily coordination and quick updates, and GitHub for version control and code collaboration. Weekly status updates are maintained through a project tracker to monitor individual contributions.

Development Environment

- **Backend Framework:** FastAPI (Python 3.12+) running via Uvicorn.
- **Frontend Framework:** Next.js 16 (React 19) with Tailwind CSS and shadcn/ui components.
- **LLM Integration:** Google Gemini API, alongside local Qwen2.5-1.5B for intent classification/term extraction, and Ollama/MedGemma as fallbacks.
- **Vector Database:** FAISS for semantic search and SapBERT embedding storage.
- **Knowledge Graph:** SNOMED CT-based structured medical knowledge representation.
- **Privacy Toolkit:** Microsoft Presidio for dynamic PII anonymization.
- **Version Control:** Git and GitHub for collaborative development.

Milestone Schedule

Milestone	Due Date	Release	Deliverable?
Identify user requirements & fix meeting with client	Jan 25, 2026	R1	No
Implement basic Chatbot session creation & GDPR consent modal	Feb 8, 2026	R1	No
Develop clinical entity extraction pipeline (Qwen2.5-1.5B)	Feb 15, 2026	R1	No
Integrate UMLS/SNOMED CT Knowledge Graph for CUI mapping	Feb 22, 2026	R1	No
R1 Presentation & Documentation Delivery	Feb 28, 2026	R1	Yes
Implement FastAPI backend & Next.js frontend interface	Mar 15, 2026	R2	No
Integrate Microsoft Presidio anonymization & advanced GDPR endpoints	Mar 25, 2026	R2	No
Implement context compaction & fallback LLM architecture	Apr 5, 2026	R2	No
Conduct system testing, RAG evaluation, and final validation	Apr 15, 2026	R2	No
Final system demo, code freeze, and project handover	Apr 26, 2026	R2	Yes